

Certified Predictive Reduction for Bounded-Memory Robot Uncertainty Monitors

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Abstract: Robotic systems increasingly rely on runtime monitors to track safety-relevant quantities computed from noisy sensor streams. Affine arithmetic gives a principled way to distinguish persistent calibration bias from fresh measurement noise, but the resulting zonotope representation grows with every measurement. Existing bounded-memory monitors therefore reduce zonotopes online, usually with a fixed geometric rule that ignores how reduction error affects future trigger verdicts. We formulate zonotope reduction as a certified abstraction-control problem: an untrusted predictive or learned policy chooses among certified reducers, while reducer certificates alone preserve soundness and the generator budget. On an omnidirectional robot monitoring benchmark, a focused receding-horizon selector improves trigger-relevant precision over a strong static Girard reducer at the same memory budget; a distilled neural selector approximates expensive decisions at lower online cost while remaining outside the trusted soundness boundary.

Keywords: Robot Monitoring, Uncertainty, Zonotopes

1 Introduction

Robots act from streams of imperfect observations. Wheel odometry, inertial measurements, range readings, and temperature or power sensors all contain uncertainty with different temporal structure: some error is persistent calibration bias, while other error is independent measurement noise. Runtime monitors are an attractive tool for such systems because they can continuously compute safety-relevant derived quantities and trigger warnings without changing the robot controller. For embedded deployment, however, monitors must also have a resource footprint that is independent of trace length.

Recent work on stream-based monitoring under uncertainty shows that affine arithmetic can track these uncertainty sources symbolically and that the affine coefficients form a zonotope [1]. This gives precise semantics for monitor state, but each fresh measurement-noise variable adds a new zonotope generator. Without approximation, memory grows forever. Classical zonotope order reduction [2, 3] solves the bounded-memory problem by replacing a high-order zonotope with a certified over-approximation containing fewer generators. The reduction choice matters: losing a correlation that appears harmless at the current time can inflate a future trigger interval after the monitor equations propagate it.

This paper treats reduction as a control decision over monitor abstractions. When the monitor state exceeds a generator budget K , a policy chooses a reducer action a from a finite certified action set, applies only the first chosen reduction, and replans at the next overflow. The policy can be static, model-predictive, or learned by imitation. It is not trusted for soundness. Soundness depends only on the reducer certificate

$$\gamma(Z) \subseteq \gamma(\rho_{a,K}(Z)), \quad \text{gen}(\rho_{a,K}(Z)) \leq K.$$

This separation lets us optimize monitor precision using approximate forecasts or neural policies while keeping the runtime-assurance argument small.

Our current evidence is a research prototype and benchmark suite for predictive zonotope reduction. The main robot benchmark is an omnidirectional monitor that tracks filtered acceleration, velocity, distance increments, position, and safety triggers under persistent calibration and fresh measurement uncertainty. At budget $K = 8$, a focused rollout selector reduces mean trigger width from 230.1 to 225.1 and interval-hull mean squared error from 352.6 to 268.4 versus the strongest static baseline, Girard. Across a budget sweep from 8 to 20 generators, the same focused selector consistently improves on Girard. The full artifact reports zero budget violations, zero unsound certificates, and zero reduction failures.

The contribution is threefold. First, we cast bounded-memory uncertain monitor execution as policy-guided certified reduction, with a policy-independent soundness argument. Second, we instantiate the idea with a focused receding-horizon selector that scores future trigger precision while preserving monitor-required calibration generators. Third, we distill expensive selector decisions into a neural policy and evaluate the current tradeoff: faster decisions and preserved certificates, but not yet a stronger precision result than focused rollout.

2 Related Work

Runtime monitoring for robotic and cyber-physical systems. Stream-based monitoring languages such as Lola and RTLola describe monitors as stream transformations whose outputs include trigger verdicts. RTLola emphasizes static analyses for bounded memory and runtime safety, and has been used in autonomous-aircraft monitoring [4, 5]. The uncertainty-aware monitor semantics of Finkbeiner et al. [1] replace exact stream values by affine forms, giving a symbolic account of calibration and measurement uncertainty. Our work keeps this semantic stack but changes the online approximation-selection step.

Zonotopes and set-based estimation. Zonotopes are widely used in reachability analysis, set-membership estimation, and fault diagnosis because they are compact and closed under linear maps and Minkowski sums [2, 6]. The practical limitation is order growth, so many reduction rules have been proposed and compared [7, 3]. These methods normally optimize present-set geometry or generic enclosure size. We instead choose among certified reductions using a monitor-aware future cost, so the objective is trigger precision after stream equations propagate the approximation.

Predictive and learned control. Our selector borrows the receding-horizon pattern from model predictive control [8]: optimize over a finite forecast, apply the first decision, and replan. The controlled variable here is not an actuator but a compression action. Learning enters as policy distillation. Imitation learning can replace expensive expert decisions with cheap policies [9]; in our setting, the learned model never justifies soundness, because every proposed action is still checked by a certified reducer.

3 Policy-Guided Certified Reduction

3.1 Monitor State

Let an uncertain monitor state be represented by a zonotope

$$Z = c + G[-1, 1]^m,$$

where c is the nominal vector of monitor stream values and each column of G is a symbolic uncertainty generator. Persistent calibration errors reuse generators across time, while fresh measurement noise appends new generators. The exact affine monitor transition is assumed sound: stepping the monitor over an uncertain state over-approximates all concrete executions represented by the input state. The difficulty is that m grows with trace length.

A bounded-memory monitor imposes a budget K on stored generators. A reducer action $\rho_{a,K}$ is admissible only when it returns a certificate that establishes both enclosure and budget:

$$\gamma(Z) \subseteq \gamma(\rho_{a,K}(Z)), \quad \text{gen}(\rho_{a,K}(Z)) \leq K. \quad (1)$$

81 The implementation includes box, Girard, Combastel, transform-based, PCA, adaptive, and scored
 82 keep-and-box reducers. A protected wrapper preserves monitor-required generators, such as calibra-
 83 tion variables, exactly before compressing the residual zonotope.

84 3.2 Policy-Independent Soundness

85 The policy is a selector

$$\pi_t : H_t \times Z_t \times \hat{x}_{t:t+h} \rightarrow A,$$

86 where H_t is observed history, Z_t is the current monitor state, $\hat{x}_{t:t+h}$ is an optional predicted input
 87 trace, and A is a finite reducer set. If Z_t exceeds budget, the monitor applies $\rho_{\pi_t, K}$. The policy may
 88 be static, predictive, learned, or suboptimal; it is sound as long as it cannot bypass Eq. (1).

89 **Proposition.** Assume the unreduced monitor transition soundly encloses all concrete states repre-
 90 sented by its input zonotope. If every applied reducer satisfies Eq. (1), then the reduced monitor
 91 execution over-approximates the unreduced affine execution at every step. Moreover, every stored
 92 post-reduction state has at most K generators.

93 The proof is by induction over monitor steps. Monitor transitions preserve sound enclosure by
 94 assumption. Reduction steps preserve enclosure by the certificate, independently of how the reducer
 95 action was selected. The budget invariant follows directly from the second part of Eq. (1).

96 3.3 Predictive and Learned Selectors

97 The focused rollout policy, MPC-G, evaluates a small protected action set for the first reduction.
 98 For each candidate it rolls the monitor forward over a short input forecast, reducing future predicted
 99 overflows with protected Girard and an emergency protected box fallback. It scores future states by
 100 trigger-relevant interval width and threshold straddling, applies only the best first action, and replans
 101 at the next real overflow.

102 We also evaluate a broader rollout ablation, MPC-W, whose first-action set includes more static
 103 protected reducers. This tests whether simply widening the action set helps. In the current artifact it
 104 does not: the broad selector is slower and less precise on the hard robot than MPC-G.

105 Finally, we train a small neural classifier from logged selector decisions. At runtime the learned
 106 policy ranks reducers, then tries them in that order until a certified budgeted reduction succeeds. If
 107 the top prediction fails the certificate or budget check, it is skipped. Thus learning affects precision
 108 and latency, not soundness.

109 4 Experiments

110 4.1 Setup

111 The evaluation uses three monitor families. The headline scenario is an omnidirectional robot monitor
 112 with persistent calibration uncertainty and fresh measurement noise. It tracks filtered acceleration,
 113 velocity, distance increments, and x/y position, then evaluates interval trigger verdicts. We also
 114 include a simpler two-axis robot monitor and a thermostat monitor as sanity checks outside the main
 115 robot setting.

116 The main run uses 50 seeds, trace length 300, generator budget $K = 8$, horizon 6, and both online
 117 and oracle predictors. Unless stated otherwise, we report online results. Metrics compare bounded
 118 monitors against an unreduced reference execution. Lower trigger width, width inflation, false-alarm
 119 rate, and interval-hull MSE are better. The unbounded reference is not a deployable bounded-memory
 120 baseline.

Table 1: Online omnidirectional robot results at budget $K = 8$, horizon 6, length 300, and 50 seeds. Lower is better except mean generators, which indicates budget usage.

Method	Trigger width	Hull MSE	False alarm	Time (s)	Mean gen.
Girard	230.1	352.6	0.0105	0.098	7.93
MPC-G	225.1	268.4	0.0113	0.957	7.93
MPC-W	242.5	728.6	0.0127	2.533	7.74
Learned	229.2	332.2	0.0115	0.169	7.92

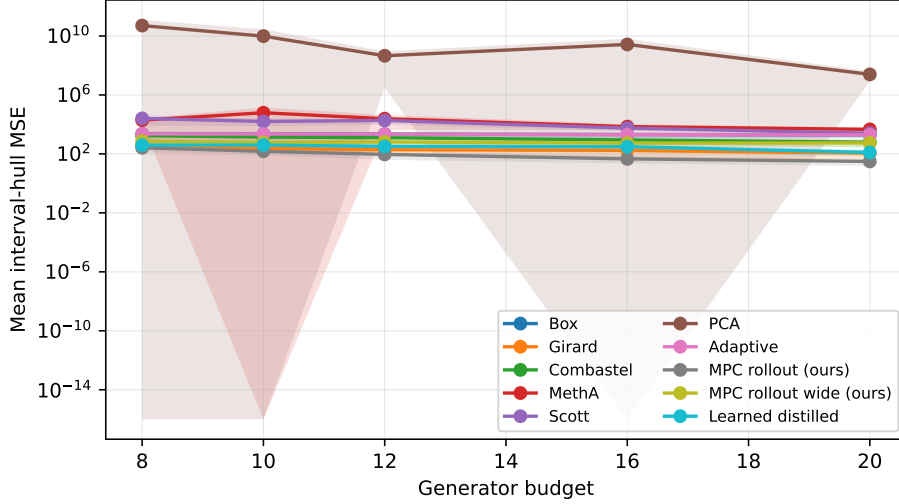


Figure 1: Omnidirectional robot interval-hull MSE by generator budget. The focused rollout selector improves over Girard across the evaluated budget range.

4.2 Hard Robot Results

Table 1 gives the main online comparison on the omnidirectional robot. Girard is the strongest static bounded-memory baseline. MPC-G improves both mean trigger width and width inflation at the same mean generator count. Its runtime is roughly ten times Girard because it performs rollout search. The learned selector is faster than rollout and close to Girard precision, but it does not match MPC-G on the main precision metric.

The budget sweep in Fig. 1 is the cleanest evidence that the predictive choice matters. MPC-G has lower interval-hull MSE than Girard for all evaluated budgets from 8 to 20 generators. The broad rollout ablation is worse despite evaluating more candidates, suggesting that monitor-aligned action design matters more than action-set size.

Figure 2 shows error over time at the default budget. The gap grows along the trace, matching the intuition that local approximation choices compound through future monitor equations.

4.3 Diagnostics and Learning

The artifact reports zero budget violations, zero unsound certificates, and zero reduction failures over 3300 aggregate run rows. This is the empirical check for the formal boundary: all state-changing decisions passed through certified reducers.

Selector diagnostics explain the method behavior. On the robot, MPC-G selects protected Girard for most reductions with occasional keep-generator choices. MPC-W uses a broader first-action distribution but pays higher decision cost and produces worse robot precision. The learned model is trained on 40,862 decision rows to imitate the broad rollout expert. Validation accuracy is 90.6% and

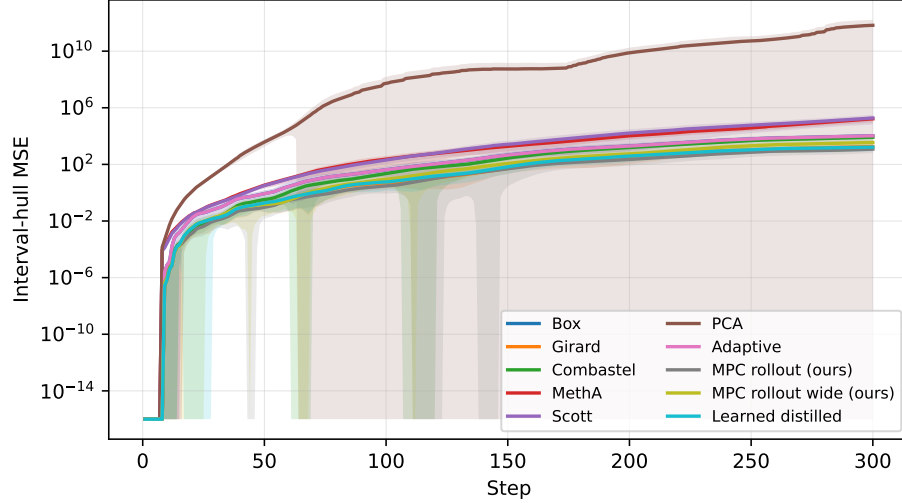


Figure 2: Ominidirectional robot error over time at the default budget.

top-3 accuracy is 98.8%, but the label distribution is highly imbalanced toward Girard. We therefore treat distillation as a safe acceleration path rather than a standalone learning result.

The simple robot and thermostat monitors are useful checks that the certified boundary and artifact pipeline work beyond one scenario. They are not the main evidence for dominance: several certified reducers saturate their trigger-width metrics near the reference.

5 Limitations

The current evidence is not yet a complete CoRL robotics story. The robot scenario is a simulation benchmark for a monitor, not a deployed robot system. Future work should evaluate on logged or live robot traces where monitor verdicts affect runtime assurance decisions. The auxiliary non-robot scenarios are too easy under the current configuration, so they mainly test generality of the implementation rather than precision improvements.

The learned selector is also preliminary. Its expert labels are imbalanced and mostly teach a Girard-heavy shortcut. A stronger learning contribution should train against the focused expert, rebalance rare actions, or use expert iteration so the policy can improve the precision/runtime frontier rather than only imitate it. Finally, the online predictor is simple; robust or tube-aware prediction may matter on traces where oracle forecasts create a larger gap.

6 Conclusion

Bounded-memory uncertainty monitors need reduction, but reduction need not be a fixed local rule. We presented policy-guided certified zonotope reduction, where predictive and learned policies optimize precision while certified reducers preserve soundness and memory bounds. Current results support a focused claim: receding-horizon reduction improves precision on a hard robot monitor at the same generator budget, and learned selectors can be placed outside the trusted soundness boundary. The next step is to turn this monitor benchmark into a robotics-facing evaluation with richer traces and stronger learned policies.

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